Daehyun Choi

Postdoctoral Researcher · Fluid Mechanics · Robotics · Bio-inspired Propulsion
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Summary

I am a postdoctoral researcher at the University of Colorado Boulder, applying fluid mechanics to vision AI, robotics, digital twins, bioinspiration, and animal behavior. My current research focuses on data-driven shape optimization of soft propulsors and on palm-scale underwater robots that breach the water surface.

Highlights

- **First-authored articles**: *J. Fluid Mech.* ($\times$3, JFM Emerging Scholar Best Paper Prize), *Phys. Fluids* (Editor's Pick), *Sci. Rep.*, *Integr. Comp. Biol.*, *Interface Focus* (co-first); 3 first-authored preprints under review (arXiv) and 2 manuscripts in preparation.
- **Awards & Grants** ($\sim$\$700k total): DARPA Young Faculty Award (\$500k, technical lead); Sejong International Postdoctoral Fellowship, NRF Korea (\$140k); Domestic Postdoctoral Scholarship, NRF Korea (\$50k); NEREM Travel Grant; IUTAM Travel Grant.

Research Interests

- **Fluid mechanics** · **Fluid-structure interaction & vortex dynamics** · **Multiphase flows** · **Bioinspired propulsion & locomotion** · **Vision AI** · **Multi-fidelity & multi-objective data-driven optimization** · **Soft robotics** · **Underwater robotics**

Skills

- **Hardware:** End-to-end device prototyping; high-speed imaging, 2D/3D PIV, 3D DIC, stereo-camera reconstruction, dynamometry; wind tunnel, water channel, pressure chamber, large-scale water tank experiments; PCB design & fabrication, circuit design, BLE wireless sensing, microcontroller data acquisition; silicone elastomer fabrication, micro-robot fabrication; biological fieldwork.
- **Software:** C++, Python, MATLAB ($>$10 years); AI-assisted development (expert level); CNN-based vision AI — DeepBubbleVelocimetry [★**9**] (developed); co-developed fluid-structure interaction simulation (OpenFOAM + CalculiX + preCICE) [arXiv]; Bayesian/multi-objective optimization (BoTorch); multiphysics simulation (ANSYS Fluent, MATLAB).
- **Leadership:** DARPA team lead, coordinating CFD (Univ. of Birmingham), experiment & optimization (Georgia Tech), and DARPA program manager; mentored $>$12 students (graduate, undergraduate, high school); peer reviewer: *J. Fluid Mech.*, *Phys. Fluids*; press: The Conversation, PBS NewsHour, Science News, & Georgia Tech News; proposal writing & grant acquisition ($\sim$\$700k); international collaboration (Korea–USA–Brunei); MSCA Seal of Excellence (96.6/100, EF-ENV).

Academic Employment

University of Colorado Boulder — Boulder, CO, USA
Postdoctoral Researcher — 2026 – present

Georgia Tech — Atlanta, GA, USA
Postdoctoral Researcher — 2023 – 2026

Ajou University — Suwon, Korea
Postdoctoral Researcher — 2022 – 2023

Education

Seoul National University — Seoul, Korea
M.S. & Ph.D. in Mechanical Engineering — 2022

Seoul National University — Seoul, Korea
B.S. in Mechanical Engineering — 2015

Publications

Bold denotes the author (Daehyun Choi). * denotes co-first authorship. † denotes co-corresponding authorship.

Journal Articles

1. **Daehyun Choi**, Kai Lauren Yung, Halley Wallace, Jacob Harrison, & S. Bhamla, "Interfacial dynamics on the water-hopping behavior of mudskipper," *In preparation.*
2. **Daehyun Choi**, Halley Wallace, Paras Singh, Chandan Bose, & S. Bhamla, "Bi-objective Bayesian optimization of soft nozzle geometry for rotary propeller," *In preparation.*
3. **Daehyun Choi***†, Ian Bergerson*, Hengjia Zhu*, Tianjun Lan, & S. Bhamla†, "Swim-and-breach at palm scale: a rudder-steered two-propeller underwater robot platform with differential-thrust pitch control," *arXiv*:2609.17240 (2026). *Under review.* arXiv
4. Paras Singh*, Yukesh Karki*, Victor Hernandez, **Daehyun Choi**, S. Bhamla, & Chandan Bose, "Multi-objective Bayesian optimization of rigid and flexible nozzles for energy-efficient pulsed jet propulsion," *arXiv*:2606.19252 (2026). *Under review.* arXiv
5. **Daehyun Choi**, Paras Singh, Ian Bergerson, Minho Kim, Jieun Park, Halley J. Wallace, Kenny Zhang, Sandy Y. Hsieh, Aqua T. Asberry, Theodore A. Uyeno, William F. Gilly, Hyungmin Park, Daeshik Kang, Chandan Bose, & S. Bhamla, "Squid funnels time their elastic recoil to amplify pulsed jets," *arXiv*:2605.03239 (2026). *Under review.* arXiv
6. **Daehyun Choi***, Paras Singh*, S. Bhamla, & Chandan Bose, "Computational and reduced-order modelling of elastic wave-driven impulse enhancement in pulsed jets through passively flexible nozzles," *arXiv*:2605.17319 (2026). *Under review.* arXiv
7. Pankaj Rohilla, Johnathan Nathaniel O'Neil, Paras Singh, Victor M. Ortega-Jimenez, **Daehyun Choi**, Chandan Bose, & S. Bhamla, "Interfacial vortex recapture enhances thrust in tiny water skaters," *bioRxiv* 2024.06.17.599397 (2025). *Under review.* bioRxiv
8. **Daehyun Choi**, Kai Yung, Ian Bergerson, Halley Wallace, Ulmar Grafe, & S. Bhamla, "Three-dimensional tracking method for water-hopping mudskippers in natural habitats," *Integr. Comp. Biol.*, 65, 1699–1713 (2025). PDF DOI
9. Pankaj Rohilla*, **Daehyun Choi***, Halley Wallace*, Kai Lauren Yung*, Juhi Deora, Atharva Lele, & S. Bhamla (*equal contribution), "Mastering the Manu—how humans create large splashes," *Interface Focus*, 15, 20240056 (2025). PDF DOI The Conversation
10. **Daehyun Choi** & Hyungmin Park, "Mechanism of enhanced impulse and entrainment of a pulsed jet through a flexible nozzle," *J. Fluid Mech.*, 996, A6 (2024). PDF DOI
11. **Daehyun Choi***, Hyunseok Kim*, & Hyungmin Park (*equal contribution), "Bubble velocimetry using the conventional and CNN-based optical flow algorithms," *Sci. Rep.*, 12, 11879 (2022). PDF DOI Code
12. **Daehyun Choi** & Hyungmin Park, "Flow–structure interaction of a starting jet through a flexible circular nozzle," *J. Fluid Mech.*, 949, A39 (2022). PDF DOI
13. **Daehyun Choi**, Jungwon Byun, & Hyungmin Park, "Analysis of liquid column atomization by annular dual-nozzle gas jet flow," *J. Fluid Mech.*, 943, A25 (2022). PDF DOI
14. Dokeun Lee, **Daehyun Choi**, Hyungmin Park, Hyomin Lee, & Sung Jae Kim, "Electroconvective circulating flows by asymmetric Coulombic force distribution in multiscale porous membrane," *J. Membr. Sci.*, 636, 119286 (2021). PDF DOI

15. **Daehyun Choi** & Hyungmin Park, "Flow around in-line sphere array at moderate Reynolds number," *Phys. Fluids* **(Editor's Pick)**, 30, 097104 (2018). PDF DOI

Conferences & Seminars

1. "Multi-fidelity optimization of a soft nozzle–propeller system," *2026 American Physical Society Global Physics Summit*, Oral presentation. Denver, USA, 2026.
2. "Hydrodynamics of water-hopping behavior of mudskippers," *2026 Society for Integrative and Comparative Biology (SICB)*, Oral presentation (Symposium). Portland, USA, 2026.
3. "Squid-inspired flexible nozzles improve thrust in rotary propulsors using a multi-fidelity approach," *2026 Society for Integrative and Comparative Biology (SICB)*, Oral presentation. Portland, USA, 2026.
4. "Squid funnel-inspired passively flexible nozzles for underwater propulsion," *2026 Society for Integrative and Comparative Biology (SICB)*, Oral presentation. Portland, USA, 2026.
5. "Geometrical effects of free-end nozzles on wave reflection in self-oscillating flexible nozzles," *78th Annual Meeting of the APS Division of Fluid Dynamics*, Oral presentation. Houston, USA, 2025.
6. "Bio-inspired flexible nozzle design for enhanced underwater propulsion," *Fluid and Biodynamics Seminar, Max Planck Institute for Dynamics and Self-Organization*, Oral presentation (Invited). Göttingen, Germany, 2025.
7. "Squid-inspired nozzles for enhanced thrust in rotary propulsors," *2025 European Mechanics Society: Bio-inspired Fluid-Structure Interaction (EUROMECH)*, Oral presentation. Vienna, Austria, 2025.
8. "Squid-inspired nozzles for enhanced thrust in rotary propulsors," *2025 American Physical Society March Meeting*, Oral presentation. Anaheim, USA, 2025.
9. "Non-invasive 3D tracking of water-hopping mudskippers in natural habitats," *2025 Society for Integrative and Comparative Biology (SICB)*, Poster presentation. Atlanta, USA, 2025.
10. "Physical modelling of water-hopping mudskipper," *77th Annual Meeting of the APS Division of Fluid Dynamics*, Poster presentation. Salt Lake City, USA, 2024.
11. "Water-hopping behavior of mudskipper," *2024 IUTAM Symposium on Capillarity and Elastocapillarity in Biology*, Poster presentation. Seoul, Korea, 2024.
12. "Hydrodynamics of water-hopping mudskipper," *2024 International Physics of Living Systems Annual Meeting*, Oral presentation. Trieste, Italy, 2024.
13. "Hydrodynamic study on the water-hopping behavior of mudskipper," *2024 American Physical Society March Meeting*, Oral presentation. Minneapolis, USA, 2024.
14. "Vortex mechanism in microvelia-inspired water-walking for thrust optimization using robotic design," *2024 Society for Integrative and Comparative Biology (SICB)*, Poster presentation. Seattle, USA, 2024.
15. "Experimental study on the starting jet through the flexible circular nozzle," *2022 World Congress on Advances in Civil, Environmental and Materials Research (ACEM)*, Oral presentation. Seoul, Korea, 2022.
16. "On the optimal flexibility condition of an impulsive starting jet nozzle for thrust generation," *73rd Annual Meeting of the APS Division of Fluid Dynamics*, Oral presentation. Online, 2020.
17. "Experimental study on the effect of nozzle flexibility on the evolution of a starting liquid jet," *72nd Annual Meeting of the APS Division of Fluid Dynamics*, Oral presentation. Seattle, USA, 2019.
18. "Study on dual-nozzle jet flows for the application of liquid atomization," *71st Annual Meeting of the APS Division of Fluid Dynamics*, Oral presentation. Atlanta, USA, 2018.
19. "Experimental study on the flow around in-line array of spheres," *69th Annual Meeting of the APS Division of Fluid Dynamics*, Oral presentation. Portland, USA, 2016.

Research Project Experience

Squid-Inspired Nozzles for Enhanced Efficiency and Thrust in Rotary Propulsors — Georgia Tech
Supported by *Defense Advanced Research Projects Agency* — 2024 – present

Hydrodynamics of Water-Hopping Mudskipper and Biomimetic Robot Development
Supported by *National Research Foundation of Korea* – Sejong International Postdoctoral Fellowship
Georgia Tech
2023 – present

Development of Cephalopod-Inspired Jet-Propelled Soft Underwater Vehicle
Supported by *National Research Foundation of Korea* – Domestic Postdoctoral Fellowship
Ajou University
2022 – 2023

Physical-Based Model for Film Thickness and Spray Characteristics Depending on Operation Conditions
Supported by *Daewoo Shipbuilding & Marine Engineering (DSME)*
Seoul National University
2021 – 2022

Apparatus for Determining Removal of Foley Catheter
Supported by *Yonsei Severance Hospital*
Seoul National University
2020 – 2022

Measurement of Bubble and Particle in Liquid
Supported by *Samsung Electro-Mechanics*
Seoul National University
2018 – 2019

Analysis of Molten Metal Atomization Using Multi-physics Simulation
Supported by *Samsung Electronics*
Seoul National University
2018 – 2018

Effects of Superhydrophobic Surface on the Wake of Circular Cylinder
Supported by *Samsung Advanced Institute of Technology (SAIT)*
Seoul National University
2016 – 2016

Awards & Grants

2025 **Emerging Scholar Best Paper Prize (Top 10),** *Journal of Fluid Mechanics*, **Cambridge University Press**
2024 **Technical Leader — DARPA Young Faculty Award ($500k), Defense Advanced Research Projects Agency**
2024 **Robert M. Nerem International Travel Award ($3k), Georgia Tech**
2024 **Young Researcher Travel Award ($1k),** IUTAM Symposium on Capillarity and Elastocapillarity in Biology
2023 **Sejong International Postdoctoral Fellowship ($140k), National Research Foundation of Korea**
2022 **Best Ph.D. Thesis Award,** Dept. of Mechanical Engineering, **Seoul National University**
2022 **Domestic Postdoctoral Scholarship ($50k), National Research Foundation of Korea**
2016 **BK21+ Doctoral Scholarship ($10k),** Ministry of Education, Korea
2015 **Presidential Scholarship for Undergraduates ($30k),** Korea
2014 **1st Prize — EDISON Challenge (CFD contest, $1k),** Ministry of Science, ICT and Future Planning, Korea

Mentoring & Teaching

2023 – present **Research Mentor,** Graduate (3) and undergraduate (5) students; 1 high school student, Georgia Tech
2022 – 2023 **Research Mentor,** 2 graduate students, Ajou University
2016 – 2022 **Undergraduate Thesis Advisor,** 3 undergraduate students, Seoul National University
2016 – 2017 **Teaching Assistant,** Mechanical Engineering Laboratory: Flow Measurements, Seoul National University

Patents (Lead Inventor)

2026 **US Patent Application, No. 19/568,682,** Compliant Nozzle System and Method for Rotary Propulsion, Georgia Tech Research Corporation

2026 **US Patent Application, No. 19/568,695,** Compliant Nozzle System and Method for Fluid Jet Impulse, Georgia Tech Research Corporation

2021 **KR Patent, 10-2021-0149171,** System for reducing insertion pressure of the ureteral access sheath for RIRS by artificially inducing hydronephrosis and measuring induced hydronephrosis with flow and hydraulic pressure, Korea

2021 **KR Patent, 10-2021-0149176,** System for reducing insertion pressure of the ureteral access sheath for RIRS by artificially inducing hydronephrosis and measuring induced hydronephrosis with hydraulic pressure and UASIF, Korea

2021 **KR Patent, 10-2021-0149175,** System for reducing insertion pressure of the ureteral access sheath for RIRS by artificially inducing hydronephrosis and measuring induced hydronephrosis with UASIF, Korea

2020 **PCT Patent, PCT/KR2020/018848,** Apparatus for determining removal of Foley catheter, International

2020 **KR Granted Patent, 10-2020-0064729,** System for reducing ureteral insertion pressure by artificially triggering hydronephrosis based on measurement of pressure associated with the insertion of the ureteral access sheath for RIRS, Korea

2020 **KR Granted Patent, 10-2020-0064731,** System for reducing ureteral entry pressure by generating vibrations in the guide wire for the insertion of the ureteral access sheath for RIRS, Korea

2020 **KR Granted Patent, 10-2020-0000288,** Apparatus for determining removal of Foley catheter, Korea